

Mathematics Imp Questions

1. **Define Communication Channel. Explain its Types.** $\Rightarrow$ A communication channel is a medium or path through which information (data or signals) is transmitted from a **sender (source)** to a **receiver (destination)**.

In digital communication, the channel may introduce **noise**, which can cause errors in transmitted data.

Types of Communication Channels

1) Noiseless Channel

A channel in which the transmitted data reaches the receiver without any error.

There is no noise or disturbance.

Ideal channel (exists only theoretically).

Example:

Data transfer inside a computer using registers.

2) Noisy Channel

A channel in which noise affects the transmitted signal.

Errors may occur during transmission.

Most real-world channels are noisy.

Examples:

Telephone lines, wireless communication, satellite communication.

3) Binary Symmetric Channel (BSC)

A special type of noisy channel.

Binary input (0 or 1).

Each bit has a probability p of being flipped.

4) Discrete Memoryless Channel (DMC)

The output depends only on the current input.

Past inputs do not affect current output.

#Define Hamming Distance. Prove that it is a Metric:-

The **Hamming distance** between two binary words (or codewords) of equal length is defined as the **number of positions at which the corresponding symbols are different**.

Let:-

$x=(x_1,x_2,\dots,x_n), y=(y_1,y_2,\dots,y_n)$ be two codewords of length n ;

Then the **Hamming distance** between x and y is denoted by $d(x,y)$ and defined as: $d(x,y)=\{i: x_i \neq y_i\}$

#Prove that:- 1) $0 \leq d(x,y) \leq n$

Proof:- • Hamming distance counts the number of positions where two codewords differ. This count can never be negative.

The maximum possible distance occurs when all n positions differ. $\Rightarrow 0 \leq d(x,y) \leq n$

2) $d(x,y)=0 \Leftrightarrow x=y$

Proof:- - If $x=y$, then all corresponding symbols are equal, so there are **no differing positions**, hence $d(x,y)=0$.

- If $d(x,y)=0$, then there are no positions where x and y differ, which implies $x=y$. **Hence - $d(x,y)=0 \Leftrightarrow x=y$**

3) Symmetry Property

$$d(x,y)=d(y,x)$$

- **Proof**

- The number of positions at which xxx differs from yyy is exactly the same as the number of positions at which yyy differs from xxx.
- Therefore, the distance is independent of order.

2. Calculate Hamming Distance Between Two Binary Words

$x=1011011$, $y=1001111$

Position	1	2	3	4	5	6	7
(x)	1	0	1	1	0	1	1
(y)	1	0	0	1	1	1	1

Different positions: 3 and 5

Total differences: 2 ans $\Rightarrow d(x,y)=2$

3. Explain Binary Symmetric Channel (BSC) with diagram

- $\Rightarrow$ The **Binary Symmetric Channel (BSC)** is a simple communication channel model used in **information theory** and **digital communication**. It is called *binary* because it only transmits two symbols (0 and 1), and *symmetric* because the probability of an error is the same in both directions.

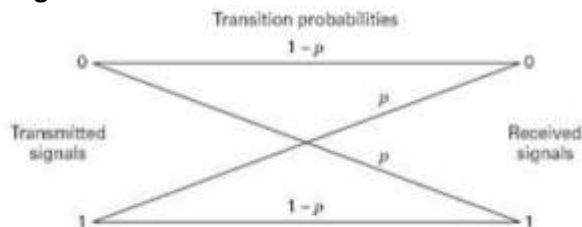
Characteristics of BSC:

The channel has **two possible input symbols**: 0 and 1.

Each transmitted bit can be **flipped** during transmission with a certain probability.

Let **p** be the probability that the bit is received incorrectly (error probability).

Diagramme:-



4. Maximum Likelihood Decoding Rule - explain with example.

$\Rightarrow$ In coding theory, when codewords are transmitted over a noisy channel, some bits may be corrupted. The **ML decoding rule** is a method to **estimate the transmitted codeword** by choosing the one that is **most likely to have produced the received vector**.

Ex:- Code: $C=\{000,111\}$ $C = \{000, 111\}$ $C=\{000,111\}$

Received vector: $y=010$ $y = 010$ $y=010$

Channel: BSC

Step (ML decoding using Hamming distance):

Distance to 000 = 1, distance to 111 = 2 $\rightarrow$ **decode as 000**

6. Nearest Neighbour Decoding Rule

Answer: **The Nearest Neighbour Decoding Rule is a method for decoding received words in coding theory.** It involves calculating the Hamming distance between the received word and all possible valid codewords. The received word is then decoded as the valid codeword that has the minimum Hamming distance from it. In simpler terms, it assumes that the transmitted codeword is the one "closest" to the received word, where "closest" is defined by the number of differing bits (Hamming distance).

Decoding a Binary Code:

Decoding a binary code means **determining the original transmitted codeword** from the received vector that may have been corrupted by errors during transmission over a noisy channel.

Steps to Decode a Binary Code

Receive the vector y from the channel.

Example: $y=1010y = 1010y=1010$

Compare with codewords in the codebook C

Suppose code $C=\{0000,1111,1010\}C = \{0000, 1111, 1010\}C=\{0000,1111,1010\}$

Choose the codeword closest to y

For a **Binary Symmetric Channel (BSC)**, use **Hamming distance**: count number of differing bits.

Decoded codeword = the codeword with **minimum Hamming distance** to y

Example: $y=1010y = 1010y=1010 \rightarrow$ distances:

$0000 \rightarrow$ distance = 2

$1111 \rightarrow$ distance = 2

$1010 \rightarrow$ distance = 0 $\rightarrow$ **decoded as 1010**

8. Error detection vs error correction

Answer: Error detection and error correction are methods used to manage errors in data transmission.

Error detection involves adding redundant information (like parity bits) to data so that the receiver can determine if any errors occurred during transmission. If an error is detected, typically a retransmission is requested.

Error correction involves adding more redundant information that not only allows the receiver to detect errors but also to identify the exact location of the errors and correct them without needing retransmission.

Error detection is simpler and requires less overhead, suitable for reliable channels where retransmission is easy. Error correction is more complex and resource-intensive, used in channels where retransmission is difficult or impossible (e.g., deep space communication).

9. Define field. State characteristics of field Q .

Answer: A field is a set F equipped with two binary operations, addition and multiplication, that satisfy the field axioms (associativity, commutativity, identity elements for both operations, additive inverses for all elements, and multiplicative inverses for all non-zero elements).

State characteristics of field $\mathbb{Q}$.

Set:

Closure:

Closed under addition, subtraction, multiplication, and division (except by zero).

Associative and Commutative:

Both addition and multiplication are associative and commutative.

Identities:

Additive identity = 0

Multiplicative identity = 1

Inverses:

Additive inverse exists for every $a \in \mathbb{Q}$: $-a$

Multiplicative inverse exists for every $a \in \mathbb{Q}, a \neq 0$: $\frac{1}{a}$

Distributive law:

Multiplication distributes over addition: $a(b+c) = ab+ac$

Therefore, $\mathbb{Q}$ satisfies **all field properties**, so it is a **field**.

What is a finite field? Explain with examples.

$\Rightarrow$ A finite field is a field that contains a finite number of elements

In other words, it is a set with a limited number of elements in which you can perform addition, subtraction, multiplication, and division (except by 0), and all the field properties are satisfied.

Characteristics of a Finite Field

It has a **finite number of elements**, say q .

It satisfies all **field axioms**: closure, associativity, commutativity, identity elements, inverses, distributivity.

Every finite field has order $q = p^n$, where:

p = a prime number (called characteristic of the field)

n = positive integer

Example: Finite field $\mathbb{Z}_5$ (integers modulo 5)

Set of elements: $\{0, 1, 2, 3, 4\}$

Addition modulo 5: $3+4 \equiv 2 \pmod{5}$

Multiplication modulo 5: $2 \cdot 3 \equiv 1 \pmod{5}$

#What is a Codeword?

A **codeword** is a sequence of symbols generated according to a **coding scheme** for transmission over a channel.

A codeword is an element of a **code**, which is a set of all codewords.

Example: In a (7,4) Hamming code, codewords are 7-bit sequences derived from 4-bit messages.

Importance:

Codewords allow the detection and correction of errors introduced during transmission.

#What is Error Probability?:- Error probability P_e is the probability that a transmitted symbol or codeword is received incorrectly. Formula: $P_e = \frac{\text{Number of errors}}{\text{Total transmissions}}$. Example: 1000 bits transmitted, 10 received incorrectly $\rightarrow P_e = \frac{10}{1000} = 0.01$. Measures reliability of communication.

8. Define Vector Space

A vector space over field F is a set V of vectors satisfying: Closure under addition and scalar multiplication, contains zero vector, satisfies associativity, commutativity, distributivity, and identity properties. Example: All n -tuples from $GF(2)$ form a vector space. Codewords of a linear code form a vector space.

9. What is a Subspace?

A subspace is a subset of a vector space that is itself a vector space. Must contain zero vector, closed under addition and scalar multiplication. Example: In $GF(2)^3$, $\{000, 111\}$ is a subspace. Linear codes are subspaces of vector space of codewords.

#Define Linear Code

A linear code C is a code where any linear combination of codewords is also a codeword. C is a subspace of $GF(q)^n$. Parameters: $[n, k, d]$ – n = codeword length, k = information bits, d = minimum Hamming distance. Example: Binary (7,4) Hamming code. Advantages: Easier encoding/decoding, efficient error detection and correction.

#Check whether a given set is a vector space over F_3

A set V is a vector space over F_3 if it satisfies:

Contains zero vector.

Closed under vector addition: $u + v \in V$ for $u, v \in V$.

Closed under scalar multiplication: $a \cdot v \in V$ for $a \in F_3$.

Satisfies associativity, commutativity, distributivity, and identity properties.

Example: $V = \{(0,0), (1,2), (2,1)\}$ over $F_3 \rightarrow$ check closure under addition and scalar multiplication. If all satisfied $\rightarrow V$ is a vector space.

#Find cyclotomic cosets modulo a given number

Cyclotomic cosets modulo n over $GF(q)$ are sets of integers $\{a, aq, aq^2, \dots\}$ modulo n until repetition occurs.

Steps:

Choose q (field size) and modulus n .

For each integer a , calculate powers of q modulo n : $a, aq, aq^2, \dots$ until it repeats modulo n .

Group numbers into distinct cosets.

Example: $q=2, n=7 \rightarrow$ cyclotomic cosets: $\{0\}, \{1,2,4\}, \{3,6,5\}$.

#Construct addition and multiplication table for $\mathbb{Z}_2[x]/(1+x+x^2)$

In $GF(2^2)$ using polynomial modulo $1+x+x^2$

Elements: $\{0, 1, x, x+1, x^2, x^2+1, x^2+x, x^2+x+1\}$, but since degree $< 2 \rightarrow \{0, 1, x, x+1\}$

Addition: Coefficients added modulo 2 (bitwise XOR).

+	0	1	x	x+1
0	0	1	x	x+1
1	1	0	x+1	x
x	x	x+1	0	1
x+1	x+1	x	1	0

Multiplication: Multiply polynomials modulo $1+x+x^2$

$\times$	0	1	x	$x+1$
0	0 0	0	0	
1	0 1	x	$x+1$	
x	0 x	$x+1$	1	
$x+1$	0 $x+1$	1	x	

18. Explain irreducible polynomials over $\mathbb{Z}_2$

=> An **irreducible polynomial** over $\mathbb{Z}_2$ is a polynomial that **cannot be factored** into polynomials of lower degree with coefficients in $\mathbb{Z}_2$.

Example: x^2+x+1 over $\mathbb{Z}_2$ is irreducible because it has no roots in $\{0,1\}$ and cannot be factored as $(x+a)(x+b)$ over $\mathbb{Z}_2$.

- **Importance:** Irreducible polynomials are used to **construct finite fields (Galois Fields)**.
- Degree n irreducible polynomial over $\mathbb{Z}_p$ gives **$\text{GF}(p^n)$** .

#Explain construction of finite field using polynomial residue class

Finite fields can be constructed using **polynomial residue class modulo an irreducible polynomial**:

Choose a prime p (field characteristic) and an irreducible polynomial $f(x)$ of degree n over $\mathbb{Z}_p$.

Elements are polynomials of degree $< n$ with coefficients in $\mathbb{Z}_p$.

Addition: Add corresponding coefficients modulo p .

Multiplication: Multiply polynomials and reduce modulo $f(x)$.

Example: $\text{GF}(2^2)$ using $f(x)=x^2+x+1$ $f(x) = x^2+x+1 \rightarrow$ elements $\{0,1,x,x+1\}$.

Addition and multiplication done modulo 2 and modulo x^2+x+1

Result: Forms a **finite field $\text{GF}(p^n)$** with p^n elements, supporting arithmetic operations.